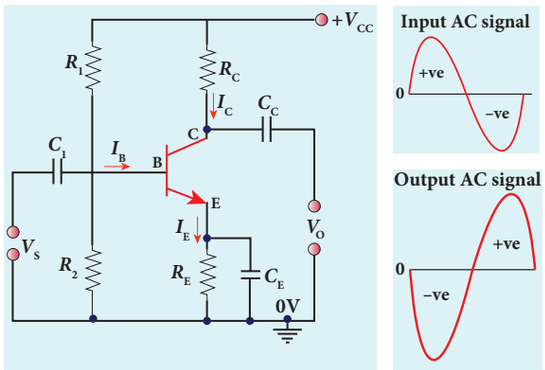


Figure 10.36 (a) Transistor as an amplifier (b) Input and output waveforms showing 180^0 phase reversal.

To start with, the Q point or the operating point of the transistor is fixed so as to get the maximum signal swing at the output (neither towards saturation point nor towards cut-off).

A load resistance R_C is connected in series with the collector circuit to measure the output voltage. The resistance R_1, R_2 and R_E form the biasing and stabilization circuit. The capacitor C_1 allows only the AC signal to pass through. The emitter bypass capacitor C_E provides a low reactance path to the amplified AC signal. The coupling capacitor C_C is used to couple one stage of the amplifier with the next stage while constructing multistage amplifiers.

V_S is the sinusoidal input signal source applied across the base-emitter. The output is taken across the collector-emitter.

$$\text{Collector current, } I_C = \beta I_B \qquad \left[\because \beta = \frac{I_C}{I_B}\right]$$

Applying Kirchhoff's voltage law to the output loop, the collector-emitter voltage is given by

$$V_{CE} = V_{CC} - I_C R_C \qquad (10.7)$$

Working of the amplifier

- *During the positive half cycle*

Input signal (V_S) increases the forward voltage across the emitter-base. As a result, the base current (I_B in μA) increases. Consequently, the collector current (I_C in mA) increases β times. This increases the voltage drop across R_c (I_cR_c) which in turn decreases the collector-emitter voltage (V_{CE}). Therefore, the input signal during the positive half cycle produces negative half cycle of the amplified signal at the output. Hence, the output signal is reversed by 180° as shown in Figure 10.36(b).

- *During the negative half cycle*

Input signal (V_S) decreases the forward voltage across the emitter-base. As a result, base current (I_B in μA) decreases and in turn decreases the collector current (I_C in mA). The decrease in collector current (I_C) decreases the potential drop across R_C which in turn increases the collector-emitter voltage (V_{CE}). Thus, the input signal during the negative half cycle produces positive half cycle of the amplified signal at the output. Therefore, 180^0 phase reversal is observed during the negative half cycle of the input signal also as shown in Figure 10.36(b).

10.4.8 Transistor as an oscillator

An electronic oscillator basically converts DC energy into AC energy of frequency ranging from a few Hz to several MHz. Hence, it is a source of alternating current or voltage. Unlike an amplifier, oscillator does not require any external signal source.

Basically, there are two types of oscillators: **Sinusoidal and non-sinusoidal**. Sinusoidal oscillators generate oscillations in the form of sine waves at constant amplitude and

XII_Physics_Vol-2_UNIT-10_Part 1_EM.indd 224 19-12-2022 16:39:11